Practice School II Report(Mid Semester) ,2026

Internship at Bitwrap Technologies

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2022BTech066

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CERTIFICATE

This certifies that Nishant Raj's Practice School-I project (2022BTech066), which he completed in partial fulfillment of the requirements for the Bachelor of Technology in Computer Science Engineering degree at JK Lakshmipat University, Jaipur, is an accurate account of the work he completed under my direction and supervision. I believe that the submitted work has met the requirements needed to be approved for the Practice School-II midterm exam

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I would like to thank my role manager**, Mr. Vipin Kumar, an Tech lead at Bitwrap Technologies**, for his unwavering assistance in comprehending and visualizing the project. His constant oversight and perceptive criticism gave the necessary guidance and clarity to complete the tasks at hand.

I would also like to convey my heartfelt gratitude to **Dr.Hanuman P Agrawal** for providing invaluable assistance whenever necessary. His guidance and encouragement have been critical to my career, and I am deeply grateful for his support

Additionally, I would like to thank the **PS-II coordination team** and **JK Lakshmipat University** for their continuous support and advice during this journey.

Sincerely yours,

Nishant Raj

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ABSTRACT

During my Practice School-II, I am working at Bitwrap Technologies under the guidance of my external mentor, **Mr. Vipin Kumar**, while **Dr. Hanuman P. Agrawal** serves as my internal faculty mentor. My work focuses on the development of a quotation automation system designed to streamline and automate the process of generating client quotations.

The system aims to reduce manual effort involved in preparing quotations by integrating structured data management, automated document generation, and workflow automation. Through this project, I am gaining practical experience in backend system development, database design, and building scalable application workflows. The project also involves designing APIs, managing data flows, and ensuring that the system can efficiently generate accurate quotations for clients with minimal manual intervention.

This experience is helping me develop a deeper understanding of building real-world software systems, backend architecture, and automation-driven product development in a professional engineering environment.

Chapter 1: ABOUT THE ORGANIZATION

1.1 Goals & Missions

The primary goal of Bitwrap Technologies is to empower businesses by leveraging modern technologies to build scalable, reliable, and innovative digital solutions. The company is committed to transforming complex business ideas into high-quality software products that drive operational efficiency and accelerate digital transformation. Its mission is to deliver cutting-edge technology solutions that enable organizations to streamline workflows, automate critical processes, and enhance their digital presence. By focusing on innovation, quality, and client-centric development, Bitwrap Technologies aims to help businesses adapt to rapidly evolving technological environments and achieve sustainable growth through technology-driven solutions.

1.2 Services & Products Offered

Bitwrap Technologies offers a comprehensive range of software development and technology services designed to help organizations build modern digital systems. The company specializes in **custom web application development**, creating scalable and high-performance platforms tailored to business needs. It also provides **mobile application development** to help organizations reach users across modern mobile ecosystems.

In addition, Bitwrap focuses on **UI/UX design**, ensuring that digital products are intuitive, visually appealing, and user-centric. The company also delivers **cloud-based solutions and system integrations**, enabling businesses to deploy scalable infrastructure and manage applications efficiently. Furthermore, Bitwrap Technologies develops **custom software solutions and automation systems** that optimize business workflows, reduce manual effort, and improve overall operational productivity.

Chapter 2: INTRODUCTION

2.1 Objective

The objective of my internship at BitWrap Technologies was to gain practical exposure to real-world software engineering practices and understand how modern software systems are designed, developed, and deployed in an industry environment. During the internship, I focused on strengthening my understanding of full-stack development by working with modern web technologies and cloud-based development tools. I learned how to design scalable software systems, structure frontend applications using component-based architectures, and build backend services that interact efficiently with databases.

Through this experience, I developed a deeper understanding of database design, API development, authentication systems, and role-based access control, as well as the integration of frontend interfaces with backend services. I also learned how to manage application state, validate data, handle file storage, and deploy applications in production environments. In addition to technical skills, the internship helped me improve important software engineering practices such as clean code organization, debugging and problem-solving, understanding product requirements, version control using Git, and collaborating within a development workflow.

Overall, the internship provided valuable experience in applying theoretical knowledge of software engineering to practical industry problems while developing the ability to design and build reliable, maintainable, and scalable software systems.

2.2 Work at Bitwrap Tech

During my internship at Bitwrap Technologies, I worked on the development of a quotation automation system designed to streamline and optimize the process of generating quotations for clients. The primary objective of this system was to reduce the manual effort involved in preparing quotations, improve accuracy, and enable faster response times for business operations. The system was designed to manage quotation data efficiently, automate document generation, and provide a structured workflow for handling client requests.

As part of my work, I was involved in the design and development of the backend architecture, where I implemented APIs to handle business logic and data processing. I also contributed to database design, structuring relational schemas to store quotation data, client information, and related business records efficiently. Additionally, I worked on integrating the frontend and backend components of the system to ensure seamless data flow and reliable system functionality.

The development process required implementing authentication and role-based access control mechanisms, ensuring that different users could securely access and manage the system based on their roles. I also worked on optimizing system workflows, automating repetitive operations, and ensuring that the application maintained clean and maintainable code structures.

Through this work, I gained practical experience in building real-world backend systems, designing scalable data models, implementing APIs, and developing automation-driven software solutions that

address real business needs. The project helped me understand how software engineering principles are applied in an industry environment to build reliable and efficient products.

Chapter 3: TOOLS & TECH STACK USED

3.1 Project Tech stack

The Ripple NDT Process Automation system was developed using a modern full-stack web development stack designed to ensure scalability, performance, and maintainability. The application follows a Single Page Application (SPA) architecture where the frontend communicates with a cloud-hosted backend and database services.

Frontend Technologies

The user interface of the application was built using **React.js**, a component-based JavaScript library that enables the development of modular and reusable UI components. The project uses **Vite** as the build tool, which provides fast development builds and optimized production bundles.

For client-side navigation, **React Router DOM** was used to implement dynamic routing across different modules such as customers, quotations, jobs, and invoicing. Application state management was handled using **Zustand**, a lightweight and efficient state management library.

Form handling was implemented using **React Hook Form**, which simplifies form validation and improves performance by reducing unnecessary re-renders. Additionally, **Zod** was used to define validation schemas for ensuring that form inputs meet the required data constraints.

The user interface styling was implemented using **Tailwind CSS**, a utility-first CSS framework that enables rapid UI development. The project also utilizes **Shadcn UI**, a component library built on **Radix UI** primitives, to create consistent and accessible interface components. Icons across the application were integrated using **Lucide React**, and date-fns was used for efficient date manipulation and formatting.

Backend and Database Technologies

The backend infrastructure of the system was built using Supabase, which provides a backend-as-a-service platform built on **PostgreSQL**. **Supabase** was used to manage database operations, authentication, file storage, and serverless API functions.

The primary database used in the system is PostgreSQL, a powerful relational database system used to store structured data related to customers, quotations, jobs, invoices, and system configurations. **Database** triggers and stored functions were implemented to automatically generate identifiers such as quotation numbers, job numbers, and invoice numbers.

Server-side logic and administrative operations were implemented using **Supabase Edge Functions**, which allow serverless execution of backend logic such as user management, role updates, and password resets.

Data Fetching and State Synchronization

For managing server data fetching and caching, the application uses **TanStack Query**, which helps handle asynchronous API calls, caching, background updates, and synchronization between the client and server.

Document Generation and File Handling

The system supports automatic generation of quotation and invoice documents using libraries such as **docx, jspdf,** and **html2canvas**, enabling the application to generate downloadable business documents directly from structured data.

File uploads and storage operations are handled through **Supabase Storage**, which securely manages uploaded files within the application.

Development Tools

The development workflow utilized tools such as **Git** for version control and Visual Studio Code as the primary development environment. These tools facilitated collaborative development, code versioning, and efficient debugging during the project lifecycle.

Overall, the technology stack used in this project enabled the development of a scalable, modular, and production-ready enterprise web application capable of automating complex business workflows in the NDT industry.

Chapter 4: ROLES AND RESPONSIBILITIES

During my internship at Bitwrap Technologies, I worked as a full-stack software developer contributing to the development of the Ripple NDT Process Automation System. My role involved participating in the design, development, and integration of different modules of the application. I was responsible for implementing features across the frontend, backend, and database layers while ensuring that the system followed clean architecture principles and scalable development practices. My work included building user interfaces, designing database structures, integrating APIs, and maintaining the overall functionality of the system.

Key Responsibilities

- Developing **frontend interfaces** using React.js and implementing modular, reusable UI components.
- Designing and implementing **database schemas and relational structures** using PostgreSQL.
- Integrating frontend applications with backend services and APIs using **Supabase**.
- Implementing **authentication and role-based access control (RBAC)** for secure system access.
- Managing application state using **Zustand** and handling form validation using **React Hook Form and Zod**.
- Building features for **customer management, quotation generation, job tracking, and invoicing modules**.
- Implementing **document generation functionality** for quotations and invoices.
- Ensuring **clean, maintainable, and well-structured code** following software engineering best practices.
- Debugging issues, optimizing system performance, and improving application workflows.
- Documenting system architecture, modules, and development processes for future maintainability.

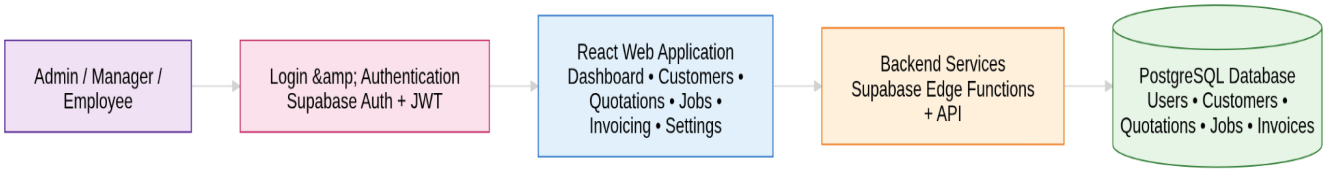
Chapter 5: WORK DONE

5.1 Internship Work Done So Far

During my internship at Bitwrap Technologies, I have contributed to the development of the Ripple NDT Process Automation System, focusing on building the foundational modules and system architecture. The work completed so far includes the following:

- Implemented the **authentication and login system** using Supabase authentication services.
- Developed **role-based access control (RBAC)** to manage permissions for different users such as administrators and employees.
- Implemented **protected routes and permission-based UI rendering** to ensure secure access to application modules.
- Designed and implemented the **database schema** for core entities including users, customers, and quotations using PostgreSQL.
- Developed the **customer management module**, enabling registration, editing, and viewing of customer information.
- Implemented features for **managing multiple contact persons and addresses** associated with customers.
- Built the initial **quotation generation workflow**, including sections such as subject, introduction, services, and tax configuration.
- Implemented **template-based quotation components** to allow reusable content and structured quotation creation.
- Developed the **core frontend architecture** using React.js with modular and reusable components.
- Implemented the **main application layout**, including navigation sidebar, top navigation, and dashboard interface.
- Integrated **state management using Zustand** for managing application state.
- Implemented **form handling and validation** using React Hook Form and Zod.
- Integrated frontend components with backend services through **Supabase database operations and API interactions**.
- Implemented **basic document generation capabilities** for creating business documents such as quotations.

These developments represent the **initial implementation phase of the system**, establishing the core architecture and functionality required for further development of advanced modules such as job management, invoicing, reporting, and workflow automation.



5.2 Week-wise-log book

Intern: Nishant Raj
Company: Bitwrap Technologies
Project: Ripple NDT Process Automation System
Duration Covered: 5 January – 6 March 2026

Date	Work Done
Jan 5 - Jan 11	Project kickoff and environment setup. Initialized React + Vite project structure. Configured Tailwind CSS and installed core dependencies (React Router, Zustand, React Hook Form, Zod). Set up Supabase project and configured authentication.
Jan 12 - Jan 18	Developed authentication system. Built Login page with Supabase Auth integration. Created ProtectedRoute and PermissionWrapper components. Implemented role-based access control (RBAC) structure with Admin/Employee roles.
Jan 19 - Jan 25	Customer Management Module - Phase 1. Built CustomerList page with search and filter functionality. Created NewCustomer form with Zod validation schema. Implemented customer contact management (Site, Account, Invoicing contacts).
Jan 26 - Feb 1	Customer Management Module - Phase 2. Developed CustomerEditModal and CustomerViewModal components. Built address management system (Registered, Site, Delivery addresses). Added GST handling and customer number auto-generation.
Feb 2 - Feb 8	Quotation System - Phase 1. Created multi-step quotation workflow structure. Built Subject and Introduction pages with template selection. Implemented Services page with dynamic pricing calculation and EditableSelectedItem component.
Feb 9 - Feb 15	Quotation System - Phase 2. Developed Taxes configuration and Commercial Terms pages. Built Technical Terms sidebar with custom categories. Created Concluding Text and Signature line components.

Feb 16 - Feb 22	Quotation System - Phase 3. Implemented GenerateQuotation page with DOCX generation using docx library. Built QuotationRegister and QuotationArchive pages. Added quotation status tracking (Draft → Sent → Accepted/Rejected).
Feb 23 - Mar 1	Jobs Module - Phase 1. Created NewJob registration form linking to quotations. Built CurrentJobs, HoldJobs, and JobReporting pages. Implemented auto-generated job numbers (JYYMMXX format). Set up job status workflow (NEW, IN_PROGRESS, HOLD, POSTPONED).
Mar 2 - Mar 6	Invoice Module - Phase 1. Designed database schema for Invoices linked to Jobs. Developed NewInvoice form with GST and tax calculations based on job data. Implemented invoice line-item management and currency handling.

Chapter 6. FUTURE PLAN

- Build InvoicingProcess page for Tax and Proforma invoice generation with auto-numbering format PI-/TI-JOB-##
- Create invoice templates with company letterhead integration and link invoices to completed jobs
- Develop PaidInvoices tracking system with payment status monitoring and Ledger view for financial tracking
- Build BankingDetails management for multiple company bank accounts and PaymentUpdates page for recording payments
- Generate invoice documents in Word/PDF format with editing capabilities and audit trail
- Build DisposedReports page for tracking physically disposed job reports and EmailReports system for managing emailed reports
- Create ArchivedJobs register with advanced search and filter functionality and JobEditor for post-completion modifications
- Implement JobRegister comprehensive view with export capabilities and job revision tracking system
- Build job cancellation and postponement workflow with reason logging and completion confirmation with digital signature
- Develop CommandLog viewer for complete audit trail of user actions and EmployeePasscode management system
- Create EmployeeList page with search, filter, sorting and EmployeeForm for adding and editing employee details
- Build role permission assignment interface with granular control and user enable/disable functionality
- Create password reset and account recovery system with session management and concurrent login control
- Create statistics cards showing total customer count, active quotations counter with status breakdown, and job metrics display
- Develop revenue tracking with monthly and quarterly views and quick action shortcuts for frequent operations
- Build notification center for system alerts and activity feed showing recent system actions
- Develop input cache system for autocomplete on frequently used fields and project name suggestions

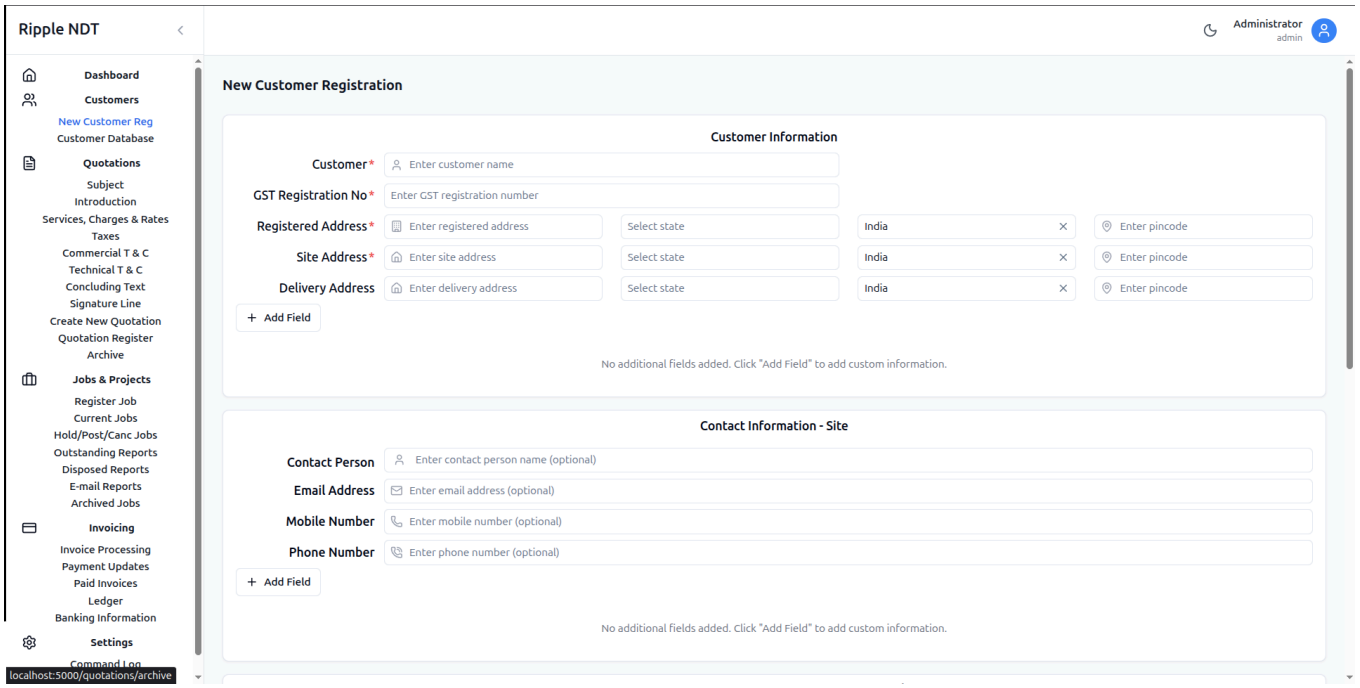
Chapter 7. CONCLUSION

The Ripple NDT Process Automation project has successfully achieved 60% completion during the period from January 5, 2026 to March 1, 2026. The foundational modules including Authentication, Customer Management, Quotation System, and initial Jobs Module have been fully developed and are ready for internal testing.

Key accomplishments include implementation of role-based access control, comprehensive customer database with contact and address management, multi-step quotation workflow with document generation, and job registration with auto-numbering capabilities. The application architecture using React, Vite, Zustand, and Supabase has proven stable and scalable for enterprise requirements.

The remaining 40% work focuses on completing the Invoicing Module, finalizing Jobs Module features including reporting and archiving, building the Settings and Administration panel, enhancing the Dashboard with analytics, and conducting thorough testing before deployment. These components will transform the application from a operational tool into a complete business management solution.

Upon full completion, this application will eliminate Excel-based workflows for Bitwrap Tech, providing a modern, secure, and efficient platform for managing NDT business processes from quotation to invoicing. The modular architecture ensures future scalability and maintenance ease.



Chapter 8. REFERENCES

React.js Documentation: Official documentation for the JavaScript library used for building user interfaces.

- [React Documentation] (https://react.dev/)

Supabase Documentation: Documentation for the open-source Firebase alternative used for backend, database, and authentication.

- [Supabase Docs] (https://supabase.com/docs)

PostgreSQL Documentation: Official documentation for the relational database system.

- [PostgreSQL Documentation] (https://www.postgresql.org/docs/)

Tailwind CSS Documentation: Utility-first CSS framework documentation.

- [Tailwind CSS Documentation] (https://tailwindcss.com/docs)

TanStack Query Documentation: Documentation for the data-fetching and state synchronization library.

- [TanStack Query] (https://tanstack.com/query/latest)

Zustand Documentation: Lightweight state management library documentation.

- [Zustand Documentation] (https://zustand-docs.pmnd.rs/)

React Hook Form Documentation: Library for form validation.

- [React Hook Form] (https://react-hook-form.com/)

Zod Documentation: TypeScript-first schema declaration and validation library.

- [Zod Documentation] (https://zod.dev/)

docx Documentation: Library used for generating Microsoft Word documents from JavaScript.

- [docx - Generating .docx files with JS/TS] (https://docx.js.org/)